Andrew W. Mann

Curriculum Vitae

PERSONAL & CONTACT INFORMATION

UNC Chapel Hill
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EDUCATION

Institute for Astronomy, University of Hawai'i at Mānoa,

Ph.D., Astronomy & Astrophysics, August 2013

PhD Thesis Topic: *Planets around cool stars: a spectroscopic and photometric study of M dwarfs and their planets*
Advisor: Prof. Eric Gaidos

M.S., Astronomy, 2010

Masters Thesis Topic: *BHOMs and the Redshift Evolution of the Cluster Merger Fraction*; Advisor: Dr. Harald Ebeling

Department of Physics and Astronomy, Johns Hopkins University,
B.S., Physics, Mathematics minor, June 2008

PROFESSIONAL EXPERIENCE

Associate Professor (with tenure), UNC Chapel Hill	2024 – Present
Assistant Professor, UNC Chapel Hill	2018 – 2024
Hubble Postdoctoral Fellow, Columbia University	2017 – 2018
Hubble Postdoctoral Fellow, UT Austin	2015 – 2017
Visiting Scientist, Boston University	2013 – 2014
Harlan J. Smith Postdoctoral Fellow, UT Austin	2013 – 2015

SELECTED HONORS

Hettleman Prize for Artistic and Scholarly Achievement, UNC	2026
NSF CAREER Award	2022
Scialog (Research Corporation for Science Advancement) Fellow	2019
NASA Hubble Fellowship	2015
Harlan J. Smith Postdoctoral Fellowship	2013
Advancing Science in America Award (ARCS)	2011

RESEARCH INTERESTS

The formation and early evolution of planetary systems, with an emphasis on the youngest planets: their detection, demographics, orbits, and atmospheres. Ages, rotation, and fundamental parameters of young and/or low-mass stars. Stellar associations. The development and implementation of statistical methods to facilitate work in these fields. Space missions and small satellites (CubeSats and SmallSats) for astrophysics.

Selected External Grants & Awards

Total external funding to UNC: $3.3M, from NASA (TESS, K2, JWST, XRP, ADAP, Keck), NSF (CAREER), and the Heising-Simons Foundation.

Teaching & Mentoring

Courses Taught

PHYS 448 Machine Learning in Physics and Astronomy (Spring 2027, upcoming; new course, co-developed with Julieta Gruszko)
ASTR 519/719 Observational Astronomy (2019 to 2026, 7 times)
ASTR 502 Modern Research in Astrophysics (Spring 2026)
ASTR 100 Understanding the Universe (Spring 2025)
ASTR 511/711 Radiative Processes in Astrophysics (Spring 2024)
ASTR 101 The Solar System (2021 to 2022, 2 times)
ASTR 202 Introduction to Astrophysics (2019 to 2020, 2 times)
PHYS 118 Introductory Mechanics and Relativity (Fall 2018)

Mentoring Summary

I have advised 10 graduate students (4 completed Ph.D.s, 1 M.S., and 5 current) and mentored 18 undergraduates (two of whom then joined my group as graduate students) and 2 high school students, 28 people in all; five current or former students have won NSF Graduate Research Fellowships, three of them held at UNC.

Current & Former Graduate Students (10)

Mia Mese (1st year, former UNC undergraduate);
PhD Topic (expected): *The activity–mass–luminosity relation*
Isabel Lopez Murillo (2nd year, former UNC undergraduate);
PhD Topic (expected): *Spot properties on young stars and high-resolution spectroscopy of planetary atmospheres*
Sarah Teague (3rd year);
PhD Topic: *(mis-?)aligned disks and planet formation*
Andrew Boyle (4th year);
PhD Topic: *Gyro-Tagging young stars with TESS*
Madyson Barber (5th year, former UNC undergraduate);
PhD Topic: *The search for young planets with TESS*
Reilly Milburn (Graduated with Ph.D. in 2025);
PhD Topic: *Photoevaporation and exospheres of young transiting planets*
Matthew Fields (Graduated with Ph.D. in 2025);
PhD Topic: *Are planets born aligned with their host stars?*
Pa Chia Thao (Graduated with Ph.D. in 2025);
PhD Topic: *Atmospheres of young exoplanets*
Jonathan Bush (Graduated with Masters in 2023);
Masters Topic: *Stellar rotation with TESS*
Mackenna Wood (Graduated with Ph.D. in 2023);
PhD topic: *Ages of young stellar associations*

Current & Former Undergraduate Students (18)

Ilakkiya Gnanakkumaran (Junior);

Project Topic: *Long-period rotation with TESS*
Lindsey Kremer (Senior);
Project Topic: *Searching for planets in Sco-Cen*
Hannah James (Senior);
Project Topic: *Transit Timing Variations in HIP 67522*
Avery Walters (Senior);
Project Topic: *Which GP Kernel is best for stellar rotation?*
Ian Sterrett (Senior);
Project Topic: *Spin-Orbit alignment of young binaries*
Leah Boff (graduated 2026);
Project Topic: *Finding False False-positive Planets*
William Storch (graduated 2026);
Project Topic: *Masses of planets from their atmospheres*
Melissa Haskell (graduated 2025);
Project Topic: *LCO follow-up of young planets*
Isabel Lopez Murillo (graduated 2025);
Project Topic: *Transit Timing Variations in Young systems*
Salem Burtner (graduated 2024);
Project Topic: *Galactic Strings, separating reality from artifacts*
Madyson Barber (graduated 2022, Undergraduate Thesis);
Project Topic: *A new young association in the Kepler field*
Stephen Schmidt (graduated 2022, Undergraduate Thesis);
Project Topic: *M dwarf metallicities from wide binaries*
Bowen Gu (graduated 2022);
Project Topic: *The role of activity on late-type dwarf stars*
Dylan Owens (graduated 2021);
Project Topic: *Eccentricities of young planets*
Patrick Gorman (graduated 2020);
Project Topic: *Design of a 1U CubeSat*
SJ Espinosa (graduated 2019);
Project Topic: *Wide Binaries in Gaia*
Jennifer Medina (TAURUS REU 2016);
Project Topic: *Rotational Broadening in Young Planet Hosts*
Richard Seifert (UT Austin; graduated 2018);
Project Topic: *Slaying Hydra (the spectrograph)*

Selected Talks and Colloquia

(Colloquium)	Michigan State University	2026
(Colloquium)	Rutgers University	2025
(Colloquium)	East Carolina University	2025
(Colloquium)	University of Michigan	2024
(Colloquium)	Johns Hopkins University	2023
(Colloquium)	Herzberg Astronomy & Astrophysics (NRC)	2021
(Invited)	*TESS* Science Team Meeting	2021
(Invited)	Sagan Summer Workshop (speaker and panelist)	2021
(Invited)	THYME conference I	2020

(Invited)	*TESS* Science Team Meeting	2020
(Invited)	UC Irvine Virtual Astronomy Series	2020
(Invited)	Kepler & K2 Science Conference V	2019
(Colloquium)	University of Hawaii at Manoa; Institute for Astronomy	2018
(Invited)	IRTF Future Directions	2018
(Colloquium)	UNC-Chapel Hill; Department of Physics and Astronomy	2018
(Colloquium)	Michigan State University; Department of Astronomy	2018
(Colloquium)	Ohio State University; Department of Astronomy	2018
(Colloquium)	University of Florida; Department of Astronomy	2018
(Invited)	Frank N. Bash Symposium; New Horizons in Astronomy	2017
(Invited)	Asteroseismology and Optical Interferometry	2017
(Colloquium)	University of Florida; Department of Astronomy	2017
(Colloquium)	Academia Sinica; Institute of Astronomy and Astrophysics	2017
(Colloquium)	University of Minnesota; Institute for Astrophysics	2017

Recent Public & Outreach Talks

Het Talks, UNC Research Week (Hettleman Prize)	2026
Saturday Science Sparks, Morehead Planetarium and Science Center	2026
Morehead Planetarium and Science Center	2025
North Carolina Museum of Natural Sciences	2024
Physics from the Ground Up, Carrboro	2023
Symposium on Horizons in Astronomy and Physics Education	2020
Astronomy on Tap, Durham	2018
Gasparilla Teachers Association	2018
Astronomy on Tap, Austin	2017
EXES Teacher Meeting	2017

Student Honors

William Storch:

NSF Graduate Research Fellowship	2026
Robert Shelton Award for Outstanding Research	2026
Paul E. Shearin Outstanding Senior Award	2026
Daniel C. Johnson Outstanding Junior Award	2025

Hannah James:

Summer Undergraduate Research Fellowship (SURF)	2026

Isabel Lopez Murillo:

Robert Shelton Award for Outstanding Research	2025
CSS Science Scholar	2023

Madyson Barber:

NSF Graduate Research Fellowship	2023
Robert Shelton Award for Outstanding Research	2022
CSS Science Scholar	2021

Andrew Boyle:

NSF Graduate Research Fellowship	2024

Stephen Schmidt:

NSF Graduate Research Fellowship	2024

Pa Chia Thao:

ExoExplorer	2023
Amelia Earhart Fellowship	2022
NSF Graduate Research Fellowship	2019–2023
Jack Kent Cooke Foundation Graduate Fellowship	2019–2022
Flatiron Predoctoral Fellowship	2021
Keck Visiting Scholar	2020

Mackenna Wood:

Three-minute thesis (3MT) winner	2022
NSF Graduate Research Fellowship, Honorable Mention	2020

Professional Service & Outreach

UNC Committees & Service

Department AI Committee (co-lead, with J. Gruszko)	2025–present
CoSMS Fundraising Committee	2025–2027
Independent Research Coordinator	2024–2027
Office of Undergraduate Research Liaison	2024–2027
Major Advising	2023–2027
SOAR science advisory board	2023–2027
Graduate Admissions & Recruiting	2018–2020
Society of Physics Students Advisor	2019–2022
Colloquia Committee	2019–2021
SALT Board of Directors	2020
Faculty Council (Natural Sciences and Mathematics); Alternate	2020

External Professional Service

Referee for Nature, the Astrophysical Journal (ApJ), the Astronomical Journal (AJ), Monthly Notices of the Royal Astronomical Society (MNRAS), Astronomy and Astrophysics (A&A), and Publications of the Astronomical Society of the Pacific (PASP).

Reviewer for NASA Postdoctoral Program (NPP)	2023–2024
Pandora Science Contributor	2022–present
Early eVolution Explorer (EVE/ESPEX) SMEX, Co-Investigator	2022–present
NASA MIDEX reviewer	
ESA's *PLATO* Target Selection Group	2020–present
NASA IRTF SPECTRE Science Advisory Committee	2019–present
Co-PI THYME Collaboration	2018–present
NASA's *TESS* Target Selection Group	2015–present
NASA's *TESS* Follow-up Group (TFOP)	2015–present
Reviewer for *JWST* and *HST*	2015, 2023, 2024
JWST proposal review panel member	
PI ZEIT Collaboration	2015–present
Reviewer for Erwin Schrödinger Fellowship	2020
NASA (XRP) external grant reviewer	2018, 2019

Public Engagement

Hidden Lives of Stars, interactive exhibit developed with the Young Worlds Lab (NSF CAREER broader impacts, co-funded by private donors), Morehead Planetarium and Science Center 2025

Refereed Publications

As of October 5, 2026, ADS lists 294 peer-reviewed publications, 20 first-author, and 22 led by UNC students. Google Scholar (as of October 5, 2026): 22,810 citations, h-index 77.
Up to date listing on Google Scholar or ADS
See publication list for more details.